$\hat{y} = mx + b$ $\qquad$ $m_i = m_0 - \alpha\left(\frac{\partial J}{\partial m}\right)$

Dataset: $(1,3)$, $(3,6)$ $\qquad$ $b_i = b_0 - \alpha\left(\frac{\partial J}{\partial b}\right)$

Initial values: $m_0 = -1$, $b_0 = 1$, $\alpha = 0.1$

Number of samples: $\cancel{n=2}$ $n = 2$

Mean Squared Error: $J(m, b) = \frac{1}{n}\sum(\hat{y} - y)^2$

Derivative of Cost Function:

Let Error: $e = \hat{y} - y$

gradient with respect to $m = \frac{\partial J}{\partial m} = \frac{1}{n}\sum 2e\frac{\partial \hat{y}}{\partial m}$

$= \frac{2}{n}\sum ex$

gradient with respect to $b = \frac{\partial J}{\partial b} = \frac{2}{n}\sum e$

Iteration 1.

Current parameter: $m = -1$, $b = 1$

Predictions for $x = 1$ $\hat{y}_1 = -1(1) + 1 = 0$

for $x = 3$ $\hat{y}_2 = -1(3) + 1 = -2$.

Errors $\delta_1 = 0 - 3 = -3$

$\delta_2 = -2 - 6 = -8$

Gradient $\frac{\partial J}{\partial m} = (-3)(1) + (-8)(3) = -27$

$\frac{\partial J}{\partial m} = -3 - 8 = -11$

Updates: $m_1 = -1 - (0.1)(-27) = 1.7$

$b_1 = 1 - 0.1(-11) = 2.1$

Iteration 2.

parameters $m = 1.7, b = 2.1$

predictions $y_1 = 1.7(1) + (2.1) = 3.8$

$y_2 = 1.7(3) + 2.1 = 7.2$

Errors: $B_1 = 3.8 - 3 = 0.8$

$B_2 = 7.2 - 6 = 1.2$

Gradients $\frac{\partial J}{\partial m} = 0.8(1) + 1.2(3) = 0.8 + 3.6$

$= 4.4$

$\frac{\partial J}{\partial b} = 0.8 + 1.2 = 2.0$

$m_2 = 1.7 - 0.1(4.4) = 1.26$

$b_2 = 2.1 - 0.1(2) = 1.0$

CS CamScanner

Iteration 3: Parameters: $m = 1.26$ $b = 1.9$

Predictions: $\hat{y}_1 = 1.26 + 1.9 = 3.16$

$\hat{y}_2 = 3(1.26) + 1.9 = 5.68$

Errors: $e_1 = 0.16$, $e_2 = -0.32$

Gradients: $\frac{\partial J}{\partial m} = 0.16(1) + (-0.32)(3)$

$= 0.16 - 0.96 = -0.8$

$\frac{\partial J}{\partial b} = 0.16 - 0.32 = -0.16$

$m_3 = 1.26 - (0.1)(-0.8)$

$= 1.34$

$b_3 = 1.9 - 0.1(-0.16) = 1.916$

Iteration 4: Parameters: $m = 1.34$, $b = 1.916$

Predictions $\hat{y}_1 = (1.34)(1) + 1.916 = 3.256$

$\hat{y}_2 = 3(1.34) + 1.916 = 5.936$

Errors: $\delta_1 = 3.256 - 3 = 0.256$

$\delta_2 = 5.936 - 6 = -0.064$

Gradients $\frac{\partial J}{\partial m} = 0.256 + (-0.064)3 = 0.064$

$\frac{\partial J}{\partial b} = 0.256 - 0.064 = 0.192$

$m_4 = 1.34 - 0.1(0.064)$ $\quad$ $b_4 = 1.916 - 0.1(0.192)$

$= 1.3336$ $\quad$ $= 1.8968$

∂m

Observations

Cost Decreasing (Evidence of Learning)

Iteration	MSE
1	36.5
2	1.04
3	0.064
4	0.0348

Error drops drastically

Trend in Parameters

Step	m	b
Start	-1	1
1	1.7	2.1
2	1.26	1.9
3	1.34	1.916
4	1.3336	1.9968

m moves from negative and stabilizes at around 1.33

b adjusts toward 1.9

Updates become smaller each step

Predictions get closer to actual y values

Error consistently decreases